Practical 15: Customer Segmentation using K-Means Clustering## Practical Information| Attribute | Details ||----------|---------|| **Practical Number** | 15 || **Title** | Customer Segmentation using K-Means Clustering || **Duration** | 4 Hours || **Course Outcomes (COs)** | CO4, CO5 || **Description** | Perform customer segmentation using clustering techniques. |## Key Topics Covered- K-Means- Elbow method- Silhouette analysis- RFM analysis- Business insights---## Instructions for Students1. **Complete all tasks** outlined in this practical2. **Write your code** in the cells provided below3. **Add comments** to explain your code4. **Include proper output** (prints, visualizations, etc.)5. **Submit your completed notebook** before the deadline---## Your Work Starts HereUse the cell(s) below to write your implementation:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

# Load built-in dataset
data = load_breast_cancer()

# Use only 2 features directly (IMPORTANT)
```

```
X = data.data[:, :2]   # first 2 features only

print("Shape of X:", X.shape)

# Scale data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
Shape of X: (569, 2)
```

```
# Apply KMeans
kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled)

print("Cluster shape:", clusters.shape)
```

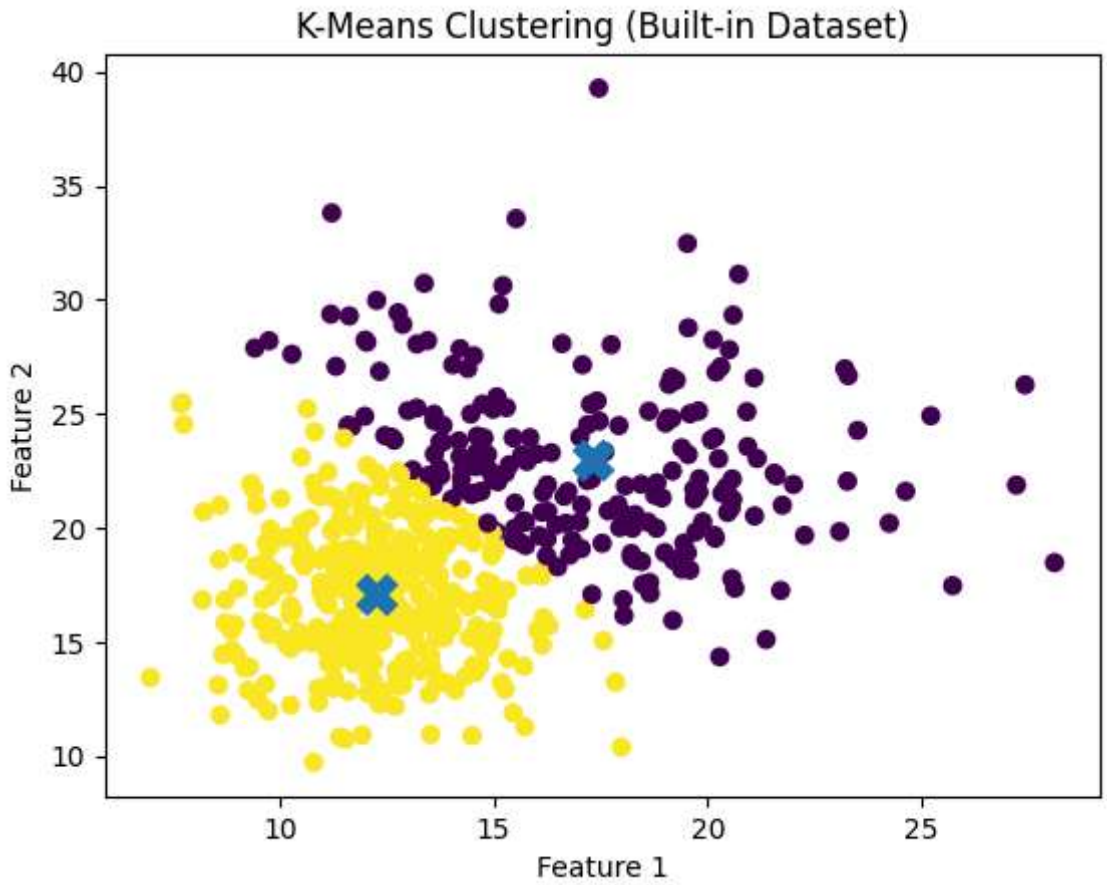
```
Cluster shape: (569,)
```

```
# Plot directly from X (no dataframe assignment)

plt.scatter(X[:,0], X[:,1], c=clusters, cmap="viridis")

# Plot centroids
centers = scaler.inverse_transform(kmeans.cluster_centers_)
plt.scatter(centers[:,0], centers[:,1], s=200, marker='X')

plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("K-Means Clustering (Built-in Dataset)")
plt.show()
```


K-Means Clustering (Built-in Dataset)
Feature 2
Feature 1
40
35
30
25
20
15
10
10
15
20
25